

BUSINESS ANALYTICS CASE STUDY

Bank Telemarketing Analytics

Customer segmentation and subscription prediction

I explored which customer groups subscribed more often and how prediction models could support campaign planning.

github.com/its-adesh/bank-telemarketing-analytics



The campaign planning question

How could the bank use customer data to prioritise outreach?



45,211

records in the analysis



I used customer groups to explore patterns, then compared models to predict subscriptions.



11.7%

subscribed to a term deposit

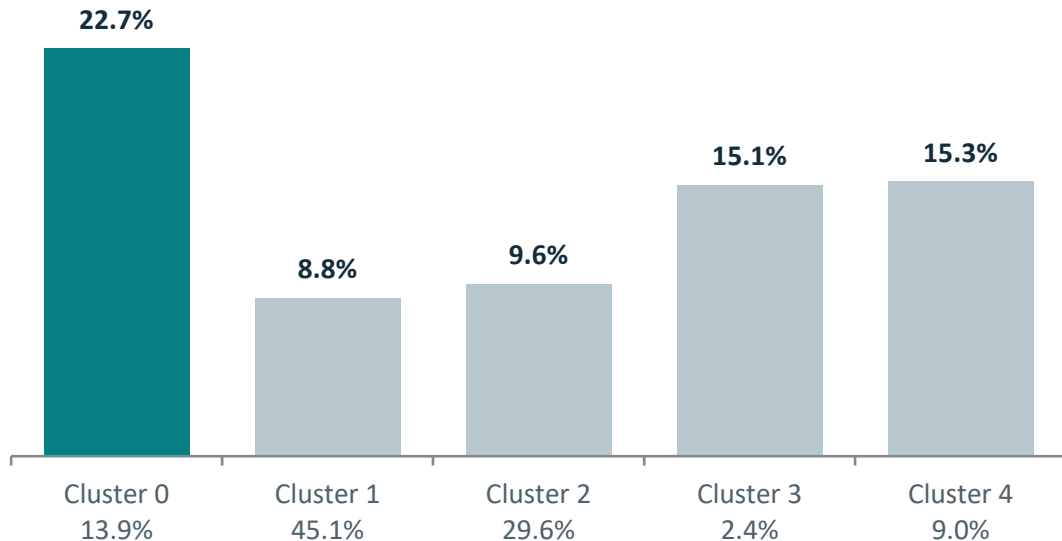


I held out 20% of the data for testing and excluded call duration from the inputs.

Cluster 0 had the highest subscription rate

The largest customer group had the lowest rate.

Subscription rate (share of customers below)



A group worth testing

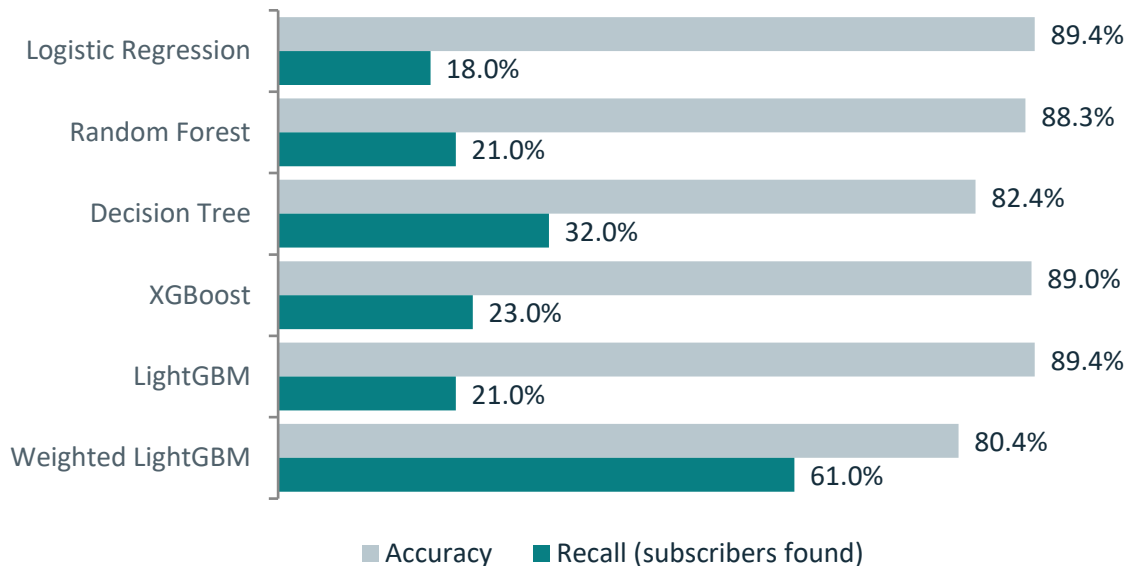
I would explore Cluster 0 in a small outreach test before expanding the campaign.



The groups overlap. A silhouette score of 0.17 supports treating them as broad profiles.

Accuracy alone missed the main problem

Predicting “no” for everyone would achieve about 88.3% accuracy.



I compared subscriber precision and recall to understand the cost of missed subscribers and incorrect predictions.

88.3%

accuracy from predicting
“no” for everyone

Weighted LightGBM

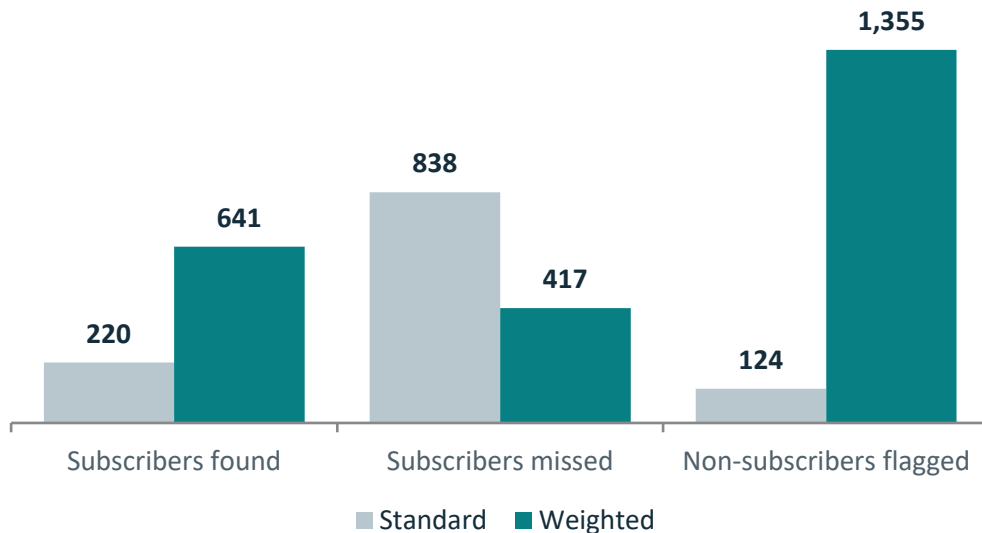
Best F1 score: 0.42

Recall 61%

Precision 32%

More subscribers found, more false positives

Class weighting changed the balance of LightGBM predictions.



Recall

21% →
61%

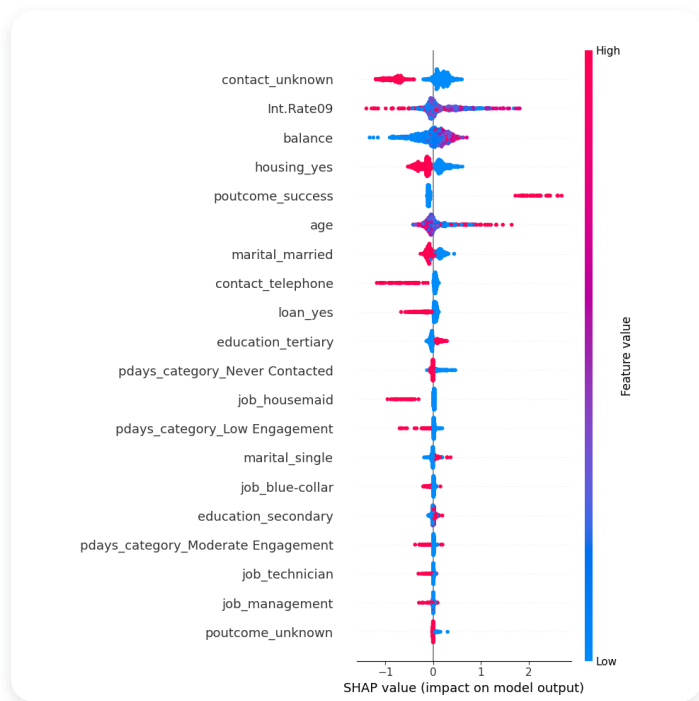
Precision

64% →
32%

I would choose the threshold around calling capacity and contact costs, then test its value in a campaign.

What influenced the predictions

SHAP helps explain the weighted LightGBM model.



How to read the chart



Each dot represents a customer record.



Right: pushes the prediction towards subscription.



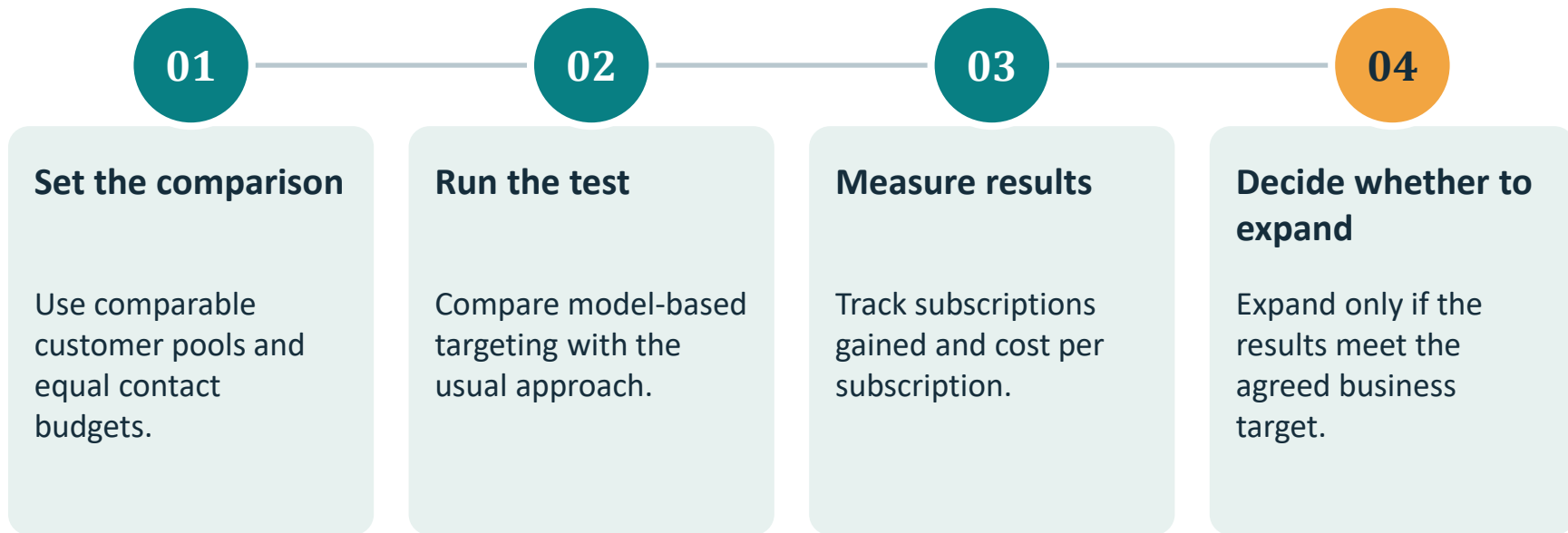
Left: pushes it away.



These patterns explain the model. They do not establish cause and effect.

A small campaign test is the next step

I recommend comparing targeted outreach with the usual approach.



My recommendation

The analysis supports a focused pilot before wider use.



Customer targeting

Explore the group with the highest historical subscription rate.



Model selection

Match the threshold to calling capacity and validate on newer data.

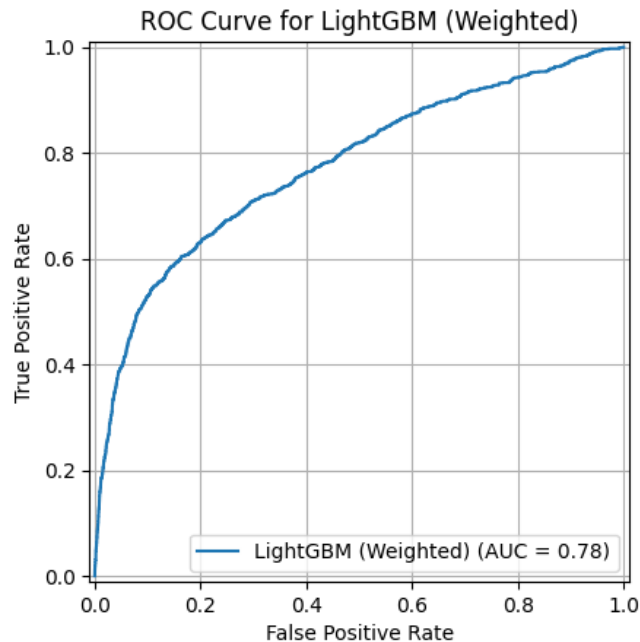


Business impact

Use the campaign test to measure value before claiming savings.

Supporting result: ROC curve

Weighted LightGBM achieved a ROC-AUC score of 0.7752.



0.7752

ROC-AUC, weighted LightGBM

A measure of ranking quality

The curve shows the balance between identifying subscribers and incorrectly flagging non-subscribers across thresholds.



The business decision still depends on contact costs and capacity.